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Application of Artificial Neural Networks and Functional Brain Connectivity to Inform Pediatric Headache

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Pediatric headache is a poorly defined disorder with unclear delineation between diagnostic sub-groups. The use of deep learning techniques, a branch of artificial intelligence (AI), has the capacity to shed light on relationships that have previously eluded investigation. We aimed to investigate the utility of artificial neural networks delineate pediatric subjects with headache from those who are otherwise healthy. We explored a series of techniques to determine the utility of AI methods in 1) compressing neuroimaging data, and 2) accurately classifying pediatric headache subjects based on neuroimaging. In a cohort of 1055 subjects from the Human Connectome Project, we trained and evaluated a variational auto-encoder (VAE), autoencoder (AE), principal component analysis (PCA), fast independent component analysis (fICA), and uniform manifold approximation and projection (UMAP) on functional magnetic resonance imaging (fMRI) data that was processed into connectivity matrices. Source reconstruction was highest for the VAE model ($R=0.828$) and lowest for UMAP ($R=0.52$) in Pearson's correlation. On an external cohort, the VAE model achieved modest reconstruction performance of around 70% accuracy for the 21 control (0.65) and 58 headache (0.63) cohorts. Feature importance analysis demonstrated the posterior division of the cingulate gyrus and its connection to the cuneal cortex to have the most remarkable impact on model accuracy. Preliminary findings support the pursuit of VAE models to understand functional neuroimaging data and should be explored to inform headache sub-groups with larger cohort sizes. NIH (R01NS125265) and Boston Children's Hospital.

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Increased Amygdala Response During Viewing of Stressful Images is Associated with Decreased Parasympathetic Tone Among Individuals with Episodic Migraine

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Migraine is a highly prevalent and debilitating headache disorder characterized by dysregulation of the autonomic nervous system (ANS) and hypersensitivity to stress. While sensitivity of the peripheral and central ANS in migraine has been demonstrated independently, simultaneous effects of stressors on both components remain unexplored. We designed a stress challenge task using images with negative and neutral valence from the International Affective Picture System (IAPS) and measured central (via fMRI amygdala activation) and peripheral (instantaneous high-frequency heart rate variability, HF-HRV, from piezoelectric pulse monitors based on (Barbieri et al., 2005) ANS reactivity in individuals with episodic migraine (MIG, $n=83$) and healthy volunteers (HV, $n=21$). Participants alternated between viewing blocks of negative- and neutral-valence images (6

images/block, 4s/image, 12 blocks/type, 144 images total, 15 min) with rest periods between blocks. For both measures, we examined the contrast between negative minus neutral blocks (NEG-NEU) and compared MIG versus HV. MIG showed greater activation during NEG-NEU in the right amygdala ($z>3.1$, $p<0.05$ FWE-corrected) and greater decreases in HF-HRV during NEG-NEU ($p=0.04$) compared to HV. Additionally, decreased HF-HRV in MIG was associated with increased left amygdala activity during NEG-NEU ($z>3.1$, $p<0.05$). These findings suggest that episodic migraine is associated with heightened central and peripheral ANS responses to stress, marked by amygdala hyperactivation and decreased parasympathetic tone. The association between decreased HF-HRV and increased amygdala reactivity underscores alterations in central autonomic regulation in response to stress in MIG patients, highlighting potential targets for interventions aimed at mitigating migraine-related stress reactivity.

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Individualized Neural Networks Differ Based on Depressive Symptoms in Women with Urologic Chronic Pelvic Pain Syndrome

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Chronic pain and depression are often comorbid, however, the neurobiological mechanisms underlying their relationship remain unknown. In this study, we examined person-specific functional connectivity in 143 women with urologic chronic pelvic pain syndrome (UCPPS) with high and low depressive symptoms (indexed by composite scores from the Hospital Anxiety and Depression Scale) from the Multidisciplinary Approach to the Study of Chronic Pelvic Pain (MAPP) Research Network. Neuroimaging included a modified "resting state" fMRI specifically aimed to evoke brain activity related to symptoms of UCPPS by instructing participants to drink water before image acquisition. Taking an idiographic analytic approach, we applied group iterative multiple model estimation (GIMME) to estimate group-, subgroup-, and individual-level connections among 12 regions of interest previously identified as key hubs for chronic pain and depression. We assessed differences in node centrality (number of contemporaneous connections involving that node, relative to the total connections for that person) between depressed and non-depressed UCPPS patients. For depressed patients, we found lower centrality in the left hippocampus and left temporal lobe, but higher centrality in the right inferior parietal lobe (all $pFDR<0.01$). Node centrality of the left temporal lobe and left thalamus predicted provoked pain sensitivity at the suprapubic region (measured by pain ratings when a 2 kg/cm² pressure was applied to the area), while the latter also predicted widespread pain (indexed by a modified CHOIR body map with 76 regions). Together, these results suggest distinct neurobiological subtypes of UCPPS based on depressive symptoms. Funding: R01DK123164.

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